

Clinical Photo Capture Guide

NEUROINJECT · SPASTICITY INJECTION GUIDE

68 muscles × 3 photos = 204 images

 Patient Position  Probe + Needle Site  Ultrasound Image

Save each photo as `<muscleId>-<type>.jpg` (type = `position · probe · us`) into `assets/images/clinical/`. JPG, landscape preferred. The probe + needle site is **one annotated photo**: draw a **blue bar** over the probe footprint on the skin and a **red dot** at the needle insertion point — save it as `<muscleId>-probe.jpg`. The app auto-detects each file and shows an "N/3 captured" counter on the muscle. **61/68** muscles are ultrasound-guided; for the rest, the US image is optional and the site photo needs only the red dot.

Pectoralis Major pec-major

Shoulder adduction, internal rotation

-  **Patient Position** pec-major-position.jpg
Position the patient supine with the arm abducted 30-45 degrees and externally rotated slightly to open the chest.
-  **Probe + Needle Site** pec-major-probe.jpg
Patient supine, arm abducted 30°. Place probe transversely on anterior chest, lateral to sternum at 3rd-4th rib. Photo from... Needle: For the clavicular head: insert the needle 2-3 cm inferior to the lateral clavicle, angled slightly caudally at about 30 degrees...
-  **Ultrasound Image** pec-major-us.jpg
Position the patient supine with the arm slightly abducted. Place the probe transversely on the anterior chest. Depth $\approx$ 5.5 cm.

Latissimus Dorsi lat-dorsi

Shoulder adduction, internal rotation, extension

-  **Patient Position** lat-dorsi-position.jpg
Position the patient in lateral decubitus with the affected side up, or prone with arm slightly abducted.
-  **Probe + Needle Site** lat-dorsi-probe.jpg
Patient lateral decubitus or prone. Place probe transversely across posterior axillary fold. Photo from behind showing probe on... Needle: Insert the needle perpendicular to the skin into the mid-belly of the muscle.
-  **Ultrasound Image** lat-dorsi-us.jpg
Identify subcutaneous tissue superficially as a hyperechoic layer.

Biceps Brachii biceps-brachii

Elbow flexion, Forearm supination

-  **Patient Position** biceps-brachii-position.jpg
Position the patient supine or seated with the arm supinated and elbow extended, resting on a support surface.
-  **Probe + Needle Site** biceps-brachii-probe.jpg
Patient supine, arm supinated. Place probe transversely on anterior mid-upper arm. Photo from lateral view showing probe across... Needle: Insert the needle into the bulk of the muscle belly at the mid-humeral level, perpendicular to the skin surface.
-  **Ultrasound Image** biceps-brachii-us.jpg
Position the patient supine with the arm slightly abducted and externally rotated. Place the probe transversely on the anterior... Depth $\approx$ 3.5-4.0 cm.

Brachialis brachialis

Pure elbow flexion

-  **Patient Position** brachialis-position.jpg
Position the patient supine or seated with the elbow slightly flexed (about 30 degrees) and the forearm supinated.
-  **Probe + Needle Site** brachialis-probe.jpg
Patient supine, elbow slightly flexed. Place probe transversely on anterior distal arm, 4-6 cm above elbow crease. Photo showing... Needle: Insert the needle perpendicular to the skin through the biceps belly, advancing into the brachialis below it.
-  **Ultrasound Image** brachialis-us.jpg
Place the probe transversely across the anterior distal third of the arm, ~4-6 cm proximal to the elbow crease.

Triceps triceps

Elbow extension

-  **Patient Position** triceps-position.jpg
Position the patient seated or prone with the arm resting on a support, elbow flexed to approximately 90 degrees.
-  **Probe + Needle Site** triceps-probe.jpg
Patient seated or prone. Place probe transversely on posterior mid-upper arm. Photo from behind showing probe across triceps... Needle: For the long head: insert the needle into the medial portion of the posterior arm at mid-humeral level, perpendicular to the skin.
-  **Ultrasound Image** triceps-us.jpg
Position the patient prone or seated with the arm supported. Place the probe transversely on the posterior arm. Depth $\approx$ 3.5-4.5 cm.

Brachioradialis brachioradialis

Elbow flexion (forearm in neutral position)

-  **Patient Position** brachioradialis-position.jpg
Position the patient seated with the forearm resting on a table in neutral position (thumb pointing upward).
-  **Probe + Needle Site** brachioradialis-probe.jpg
Patient seated, forearm neutral on table. Place probe transversely on lateral proximal forearm, 4-6 cm below lateral epicondyle.... Needle: Insert the needle into the bulk of the muscle belly at the junction of the proximal and middle thirds of the forearm,...
-  **Ultrasound Image** brachioradialis-us.jpg
Place the probe transversely on the lateral forearm with the patient's elbow slightly flexed and forearm in neutral (thumb-up)... Depth $\approx$ 3.5 cm.

Pronator Teres pronator-teres

Forearm pronation

-  **Patient Position** pronator-teres-position.jpg
Position the patient seated with the forearm supinated and the elbow extended, resting on a support surface.
-  **Probe + Needle Site** pronator-teres-probe.jpg
Patient seated, forearm supinated. Place probe transversely on medial proximal forearm, 3-5 cm below medial epicondyle. Photo... Needle: Insert the needle 2-3 cm distal to the medial epicondyle, into the bulk of the muscle belly on the medial-to-anterior proximal...
-  **Ultrasound Image** pronator-teres-us.jpg
Position the patient with the elbow extended and forearm supinated to stretch the pronator teres. Depth $\approx$ 4.5 cm.

Flexor Carpi Radialis fcr

Wrist flexion, Radial deviation

-  **Patient Position** fcr-position.jpg
Position the patient seated with the forearm supinated and resting on a flat surface.
-  **Probe + Needle Site** fcr-probe.jpg
Patient seated, forearm supinated on table. Place probe transversely on central volar forearm at proximal-to-mid third. Photo... Needle: Insert the needle into the bulk of the muscle belly at the junction of the proximal and middle thirds of the forearm,...
-  **Ultrasound Image** fcr-us.jpg
Place the probe transversely on the volar forearm, centered over the palpable FCR tendon. Depth $\approx$ 2.0 cm.

Flexor Carpi Ulnaris fcu

Wrist flexion, Ulnar deviation

-  **Patient Position** fcu-position.jpg
Position the patient seated with the forearm supinated or in neutral, resting on a support.
-  **Probe + Needle Site** fcu-probe.jpg
Patient seated, forearm supinated. Place probe transversely on medial forearm. Photo showing probe on ulnar/medial side of... Needle: Insert the needle into the bulk of the muscle belly at the junction of the proximal and middle thirds of the forearm, on the...
-  **Ultrasound Image** fcu-us.jpg
Place the probe transversely on the medial (ulnar) side of the forearm with the forearm fully supinated. Depth $\approx$ 2.5-3.5 cm.

Flexor Digitorum Superficialis fds

Finger flexion at PIP joints, Wrist flexion

☒ Patient Position fds-position.jpg

Position the patient seated with the forearm supinated and resting on a flat surface.

☐ Probe + Needle Site fds-probe.jpg

Patient seated, forearm supinated on table. Place probe transversely on volar forearm, 5-8 cm below medial epicondyle. Photo... Needle: Insert the needle at the mid-forearm level, between the FCR and FCU, slightly deeper than the superficial flexor layer.

☐ Ultrasound Image fds-us.jpg

Place the probe transversely on the central volar forearm. Depth $\approx$ 2.0-3.0 cm.

Flexor Digitorum Profundus fdp

Finger flexion at DIP joints, Wrist flexion

☒ Patient Position fdp-position.jpg

Position the patient seated with the forearm supinated and resting on a flat surface.

☐ Probe + Needle Site fdp-probe.jpg

Patient seated, forearm supinated. Place probe transversely on medial-volar mid-forearm. Photo showing probe on deep volar... Needle: Insert the needle from the medial (ulnar) side of the forearm, just anterior to the palpable subcutaneous ulnar border, at the...

☐ Ultrasound Image fdp-us.jpg

Place the probe transversely on the volar-ulnar forearm at the mid-forearm level. Depth $\approx$ 3.0-4.0 cm.

Flexor Pollicis Longus fpl

Thumb flexion at IP joint

☒ Patient Position fpl-position.jpg

Position the patient seated with the forearm supinated and resting flat on a support surface.

☐ Probe + Needle Site fpl-probe.jpg

Patient seated, forearm supinated flat. Place probe transversely on mid-volar forearm at junction of middle and distal thirds.... Needle: Insert the needle at the mid-forearm level, on the radial side, between the brachioradialis (laterally) and the FCR (medially).

☐ Ultrasound Image fpl-us.jpg

Place the probe transversely on the volar forearm with the patient supinated. Depth $\approx$ 3.0-3.5 cm.

Adductor Pollicis adductor-pollicis

Thumb adduction (thumb-in-palm deformity)

☒ Patient Position adductor-pollicis-position.jpg

Position the patient with the hand resting on a flat surface, palm down (pronated), with the thumb abducted away from the palm.

☐ Probe + Needle Site adductor-pollicis-probe.jpg

Patient hand pronated on table. Place small linear or hockey-stick probe transversely across first web space from dorsal... Needle: Insert the needle from the dorsal aspect of the first web space, perpendicular to the plane of the hand.

☐ Ultrasound Image adductor-pollicis-us.jpg

Identify the subcutaneous fat and thin dorsal skin as the most superficial layers.

Subscapularis subscapularis

Shoulder internal rotation, Adduction

☒ Patient Position subscapularis-position.jpg

Position the patient supine or seated with the arm abducted to 45-60 degrees and externally rotated, resting on a support.

☐ Probe + Needle Site subscapularis-probe.jpg

Patient supine, arm abducted 45-60° and externally rotated. Place probe parasagittally in anterior axilla. Photo from front... Needle: Insert the needle through the ANTERIOR axillary fold (formed by pec major), directing it posterolaterally toward the anterior...

☐ Ultrasound Image subscapularis-us.jpg

Identify the subcutaneous fat and the pectoralis major as the most superficial muscular layer on the anterior chest.

Infraspinatus infraspinatus

 NOT TYPICALLY INJECTED — External rotator (reference only)

☒ Patient Position infraspinatus-position.jpg

Position the patient seated, leaning forward, or in a prone position with the arms at the sides.

☐ Probe + Needle Site infraspinatus-probe.jpg

Patient seated leaning forward or prone. Place probe transversely across infraspinous fossa, parallel to scapular spine. Photo... Needle: Insert the needle perpendicular to the skin into the center of the infraspinous fossa, targeting the thickest part of the muscle...

☐ Ultrasound Image infraspinatus-us.jpg

Identify the subcutaneous fat and the thin trapezius muscle as the most superficial layers (the trapezius overlies the...

Teres Major teres-major

Shoulder adduction, Internal rotation, Extension

☒ Patient Position teres-major-position.jpg

Position the patient seated or prone with the arm adducted at the side.

☐ Probe + Needle Site teres-major-probe.jpg

Patient seated or prone, arm at side. Place probe transversely over posterior axillary fold. Photo from behind showing probe on... Needle: Identify the posterior axillary fold and the inferior angle of the scapula.

☐ Ultrasound Image teres-major-us.jpg

Position the patient seated or prone with the arm slightly abducted. Place the probe on the posterior axillary fold. Depth $\approx$ 3.5-4.5 cm.

Pronator Quadratus pronator-quadratus

Forearm pronation (distal)

☒ Patient Position pronator-quadratus-position.jpg

Position the patient supine or seated with the forearm supinated and resting on a support, wrist in neutral.

☐ Probe + Needle Site pronator-quadratus-probe.jpg

Patient supine, forearm supinated on table. Place probe transversely on volar distal forearm, 3-5 cm proximal to wrist crease.... Needle: DORSAL APPROACH (preferred): Insert the needle from the dorsal distal forearm, between the radius and ulna, advancing through the...

☐ Ultrasound Image pronator-quadratus-us.jpg

Place the probe transversely on the volar distal forearm, 3-5 cm proximal to the wrist crease.

Hand Lumbricals hand-lumbricals

MCP flexion + IP extension (intrinsic-plus posture)

☒ Patient Position hand-lumbricals-position.jpg

Position the patient with the hand supinated (palm up), fingers relaxed and slightly extended.

☐ Probe + Needle Site hand-lumbricals-probe.jpg

Patient palm-up, fingers relaxed. Place hockey-stick probe longitudinally on palm along the metacarpal. Photo from palmar view... Needle: PALMAR APPROACH: Insert the needle into the palm just proximal to the MCP joint of the target finger, on the radial side of the...

☐ Ultrasound Image hand-lumbricals-us.jpg

Place the probe longitudinally on the palmar surface along the metacarpal of the target finger.

Quadratus Lumborum quadratus-lumborum

Pelvic hiking, trunk lateral flexion

-  **Patient Position** [quadratus-lumborum-position.jpg](#)
Position the patient prone with a pillow under the abdomen to reduce lumbar lordosis.
-  **Probe + Needle Site** [quadratus-lumborum-probe.jpg](#)
Patient prone, pillow under abdomen. Place curvilinear probe transversely on flank, just above iliac crest. Photo from side... Needle: Insert the needle perpendicular to the skin, angling slightly medially toward the transverse processes of L3/L4.
-  **Ultrasound Image** [quadratus-lumborum-us.jpg](#)
Identify the subcutaneous fat and thoracolumbar fascia superficially as hyperechoic layers.

Rectus Abdominis rectus-abdominis

Trunk flexion (camptocormia / forward-flexed posture)

-  **Patient Position** [rectus-abdominis-position.jpg](#)
Position the patient supine with the knees slightly bent to relax the abdominal wall.
-  **Probe + Needle Site** [rectus-abdominis-probe.jpg](#)
Patient supine, knees bent. Place linear probe transversely on the anterior abdomen, just lateral to midline at the umbilical... Needle: Inject at 2-4 sites distributed along the length of the muscle — typically at the supra-umbilical and infra-umbilical levels...
-  **Ultrasound Image** [rectus-abdominis-us.jpg](#)
Place the probe transversely on the anterior abdominal wall, just lateral to the midline at the level of the umbilicus.

Obliques (External & Internal) obliques

Trunk rotation, lateral flexion

-  **Patient Position** [obliques-position.jpg](#)
Position the patient supine or in the lateral decubitus position with the target side up.
-  **Probe + Needle Site** [obliques-probe.jpg](#)
Patient supine, knees bent. Place linear probe transversely on the lateral abdominal wall between the costal margin and... Needle: Target the external oblique and/or internal oblique depending on the clinical pattern — typically inject the internal oblique as...
-  **Ultrasound Image** [obliques-us.jpg](#)
Place the probe transversely on the lateral abdominal wall at the level of the umbilicus, along the mid-axillary line.

Paraspinals (Erector Spinae Group) paraspinals

Trunk extension spasticity, axial dystonia, camptocormia treatment

-  **Patient Position** [paraspinals-position.jpg](#)
Position the patient prone with a pillow under the abdomen to reduce lumbar lordosis and open the interlaminar spaces.
-  **Probe + Needle Site** [paraspinals-probe.jpg](#)
Patient prone, pillow under abdomen. Place curvilinear probe parasagittally 2-3 cm lateral to midline spinous processes, or... Needle: Insert the needle perpendicular to the skin or at a slight lateral-to-medial angle, advancing into the muscle belly.
-  **Ultrasound Image** [paraspinals-us.jpg](#)
PARASAGITTAL VIEW: Place the probe longitudinally (parasagittal), 2-3 cm lateral to the midline spinous processes. The transverse...

Gluteus Maximus gluteus-maximus

Hip extension, external rotation

-  **Patient Position** [gluteus-maximus-position.jpg](#)
Position the patient prone or in lateral decubitus with the affected side up.
-  **Probe + Needle Site** [gluteus-maximus-probe.jpg](#)
Patient prone. Place curvilinear probe transversely on upper-outer quadrant of buttock. Photo from behind showing probe position... Needle: Insert the needle perpendicular to the skin into the muscle bulk at multiple sites distributed across the belly (3-4 sites for a...)
-  **Ultrasound Image** [gluteus-maximus-us.jpg](#)
Identify the thick subcutaneous fat layer, which can be substantial in this region.

Piriformis piriformis

Hip external rotation, abduction

-  **Patient Position** [piriformis-position.jpg](#)
Position the patient prone with a pillow under the pelvis to slightly flex the hips.
-  **Probe + Needle Site** [piriformis-probe.jpg](#)
Patient prone, pillow under pelvis. Place curvilinear probe transversely along PSIS-to-greater-trochanter line, at midpoint.... Needle: Insert the needle perpendicular to the skin surface at the midpoint.
-  **Ultrasound Image** [piriformis-us.jpg](#)
Identify the subcutaneous tissue and the thick Gluteus Maximus as a large superficial hyperechoic muscle.

Tensor Fasciae Latae (TFL) tfl

Hip flexion, abduction, internal rotation

-  **Patient Position** [tfl-position.jpg](#)
Position the patient supine or in lateral decubitus with the affected side up.
-  **Probe + Needle Site** [tfl-probe.jpg](#)
Patient supine or lateral. Place probe transversely on anterolateral hip, 2-3 cm below ASIS. Photo showing probe just distal to... Needle: Insert the needle perpendicular to the skin surface directly into the palpated muscle belly.
-  **Ultrasound Image** [tfl-us.jpg](#)
Identify the subcutaneous fat as a superficial hyperechoic layer — it is usually thin in this region.

Rectus Femoris rectus-femoris

Stiff knee gait, Knee extension

-  **Patient Position** [rectus-femoris-position.jpg](#)
Position the patient supine with the knee extended and the thigh relaxed.
-  **Probe + Needle Site** [rectus-femoris-probe.jpg](#)
Patient supine, knee extended. Place probe transversely on anterior mid-thigh. Photo from lateral view showing probe across... Needle: Insert the needle perpendicular to the skin into the bulk of the muscle belly at mid-thigh.
-  **Ultrasound Image** [rectus-femoris-us.jpg](#)
Identify subcutaneous fat and the superficial fascia (fascia lata) as the most superficial hyperechoic layers.

Vastus Medialis vastus-medialis

Knee extension (distal stabilization)

-  **Patient Position** [vastus-medialis-position.jpg](#)
Position the patient supine with the knee extended and the thigh relaxed.
-  **Probe + Needle Site** [vastus-medialis-probe.jpg](#)
Patient supine, knee extended. Place probe transversely on distal medial thigh, just above patella (VMO area). Photo showing... Needle: Insert the needle perpendicular to the skin into the bulk of the VMO muscle belly.
-  **Ultrasound Image** [vastus-medialis-us.jpg](#)
Identify subcutaneous tissue and the superficial fascia as hyperechoic layers.

Vastus Lateralis vastus-lateralis

Knee extension

-  **Patient Position** [vastus-lateralis-position.jpg](#)
Position the patient supine or in lateral decubitus with the affected side up.
-  **Probe + Needle Site** [vastus-lateralis-probe.jpg](#)
Patient supine or lateral. Place probe transversely on lateral mid-thigh. Photo from lateral view showing probe on outer thigh. Needle: Insert the needle through the IT band, angling slightly anteriorly to enter the VL muscle belly.
-  **Ultrasound Image** [vastus-lateralis-us.jpg](#)
Identify subcutaneous fat and the fascia lata (including the IT band component) as hyperechoic superficial layers.

Vastus Intermedius vastus-intermedius

Knee extension

 **Patient Position** vastus-intermedius-position.jpg
Position the patient supine with the knee extended and the thigh relaxed.

 **Probe + Needle Site** vastus-intermedius-probe.jpg
Patient supine, knee extended. Place probe transversely on anterior proximal thigh (same region as rectus femoris but deeper... Needle: Insert the needle perpendicular to the skin through the center of the Rectus Femoris.

 **Ultrasound Image** vastus-intermedius-us.jpg
Identify subcutaneous fat and fascia lata superficially.

Semimembranosus semimembranosus

Knee flexion

 **Patient Position** semimembranosus-position.jpg
Position the patient prone with the knee slightly flexed (place a pillow under the ankle).

 **Probe + Needle Site** semimembranosus-probe.jpg
Patient prone, knee slightly flexed. Place probe transversely on posterior medial thigh at mid-to-distal level. Photo from behind... Needle: Insert the needle from the medial aspect, angling slightly posteriorly and laterally to pass deep to the Semitendinosus.

 **Ultrasound Image** semimembranosus-us.jpg
Identify subcutaneous tissue and the deep fascia as superficial hyperechoic layers.

Semitendinosus semitendinosus

Knee flexion

 **Patient Position** semitendinosus-position.jpg
Position the patient prone with the knee slightly flexed (place a pillow under the ankle).

 **Probe + Needle Site** semitendinosus-probe.jpg
Patient prone, knee slightly flexed. Place probe transversely on posterior mid-thigh. Photo from behind showing probe on... Needle: Insert the needle perpendicular to the skin into the mid-belly of the muscle (proximal to the tendinous portion).

 **Ultrasound Image** semitendinosus-us.jpg
Identify subcutaneous tissue and the deep fascia superficially.

Gastrocnemius (Medial) gastrocnemius-medial

Equinus deformity, Plantarflexion

 **Patient Position** gastrocnemius-medial-position.jpg
Position the patient prone with the feet hanging off the edge of the bed.

 **Probe + Needle Site** gastrocnemius-medial-probe.jpg
Patient prone, feet off bed edge. Place probe transversely on proximal medial calf, 2-3 cm below popliteal crease. Photo from... Needle: Insert the needle perpendicular to the skin into the bulk of the medial gastrocnemius belly.

 **Ultrasound Image** gastrocnemius-medial-us.jpg
Identify subcutaneous fat and the superficial fascia.

Gastrocnemius (Lateral) gastrocnemius-lateral

Equinus deformity, Plantarflexion

 **Patient Position** gastrocnemius-lateral-position.jpg
Position the patient prone with the feet hanging off the edge of the bed.

 **Probe + Needle Site** gastrocnemius-lateral-probe.jpg
Patient prone, feet off bed edge. Place probe transversely on proximal lateral calf, 2-3 cm below popliteal crease. Photo from... Needle: Insert the needle perpendicular to the skin into the lateral muscle belly.

 **Ultrasound Image** gastrocnemius-lateral-us.jpg
Identify subcutaneous fat and superficial fascia.

Soleus (Medial) soleus-medial

Plantarflexion

 **Patient Position** soleus-medial-position.jpg
Position the patient prone with the feet hanging off the edge of the bed.

 **Probe + Needle Site** soleus-medial-probe.jpg
Patient prone, feet off edge. Place probe transversely on medial mid-calf, just posterior to tibial border. Photo from... Needle: Insert the needle perpendicular to the skin, directed slightly laterally and posteriorly.

 **Ultrasound Image** soleus-medial-us.jpg
Identify subcutaneous tissue and superficial fascia.

Soleus (Lateral) soleus-lateral

Plantarflexion

 **Patient Position** soleus-lateral-position.jpg
Position the patient prone with the feet hanging off the edge of the bed.

 **Probe + Needle Site** soleus-lateral-probe.jpg
Patient prone, feet off edge. Place probe transversely on lateral mid-calf. Photo from behind showing probe on lateral mid-calf. Needle: Insert the needle perpendicular to the skin into this lateral soleus bulge at mid-calf level.

 **Ultrasound Image** soleus-lateral-us.jpg
Identify subcutaneous tissue and superficial fascia.

Flexor Digitorum Longus (FDL) fdl

Claw toes, toe flexion

 **Patient Position** fdl-position.jpg
Position the patient prone or supine with the leg externally rotated to expose the medial calf.

 **Probe + Needle Site** fdl-probe.jpg
Patient prone or supine with leg externally rotated. Place probe transversely on medial distal leg, just behind tibial border.... Needle: Insert the needle perpendicular to the skin, directing it toward the posterior aspect of the tibia and fibula.

 **Ultrasound Image** fdl-us.jpg
Identify subcutaneous tissue and the superficial crural fascia.

Flexor Digitorum Brevis (FDB) fdb

Toe flexion (plantar)

 **Patient Position** fdb-position.jpg
Position the patient prone with the foot hanging off the edge of the bed, or supine with the ankle dorsiflexed to expose the sole.

 **Probe + Needle Site** fdb-probe.jpg
Patient prone, foot hanging off bed. Place probe longitudinally on plantar midfoot from calcaneus toward toes. Photo from plantar... Needle: Insert the needle perpendicular to the plantar skin surface.

 **Ultrasound Image** fdb-us.jpg
Identify the thick hyperechoic plantar fascia as the most superficial fibrous layer on the sole.

Tibialis Anterior tibialis-anterior

⚠️ NOT TYPICALLY INJECTED — Dorsiflexor (reference only)

 **Patient Position** tibialis-anterior-position.jpg
Position the patient supine with the leg extended and the ankle in neutral.

 **Probe + Needle Site** tibialis-anterior-probe.jpg
Patient supine, leg extended. Place probe transversely on proximal anterolateral leg, 2 cm lateral to tibial crest. Photo from... Needle: Insert the needle perpendicular to the skin into the largest muscle bulk, approximately 2 cm lateral to the tibial crest.

 **Ultrasound Image** tibialis-anterior-us.jpg
Identify subcutaneous tissue — it is thin on the anterior leg.

Tibialis Posterior tibialis-posterior

Equinovarus (Plantarflexion + Inversion)

 **Patient Position** tibialis-posterior-position.jpg
Position the patient prone or supine with the leg externally rotated.

 **Probe + Needle Site** tibialis-posterior-probe.jpg
Patient prone or supine with leg externally rotated. Place probe transversely on medial mid-leg, just behind tibia. Photo from... Needle: Insert the needle directed posterolaterally, sliding along the posterior aspect of the tibia.

 **Ultrasound Image** tibialis-posterior-us.jpg
Identify subcutaneous tissue and the crural fascia superficially.

Flexor Hallucis Longus (FHL) fh1

Claw hallux, Equinovarus support

 **Patient Position** fh1-position.jpg
Position the patient prone with the feet hanging off the edge of the bed.

 **Probe + Needle Site** fh1-probe.jpg
Patient prone, feet off edge. Place probe transversely on distal posterolateral calf, adjacent to fibula. Photo from behind... Needle: Insert the needle lateral to the Achilles tendon, directed toward the posterior fibula.

 **Ultrasound Image** fh1-us.jpg
Identify subcutaneous tissue and the Soleus muscle/aponeurosis superficially.

Extensor Hallucis Longus (EHL) eh1

Hallux extension, hitchhiking thumb sign

 **Patient Position** eh1-position.jpg
Position the patient supine with the leg extended and the ankle in neutral.

 **Probe + Needle Site** eh1-probe.jpg
Patient supine, leg extended. Place probe transversely on distal anterolateral leg, ~10 cm above ankle. Photo from front showing... Needle: Insert the needle perpendicular to the skin into the EHL muscle belly.

 **Ultrasound Image** eh1-us.jpg
Identify the thin subcutaneous tissue on the anterior leg.

Adductor Hallucis add-hallucis

Hallux adduction, transverse arch collapse

 **Patient Position** add-hallucis-position.jpg
Position the patient supine with the foot accessible, or prone with the foot dorsiflexed.

 **Probe + Needle Site** add-hallucis-probe.jpg
Patient supine, foot accessible. Place probe longitudinally along 1st-2nd intermetatarsal space on plantar surface. Photo showing... Needle: A dorsal approach is an alternative: insert the needle from the dorsum of the foot between the 1st and 2nd metatarsal shafts,...

 **Ultrasound Image** add-hallucis-us.jpg
Identify the hyperechoic cortex of the 1st and 2nd metatarsal shafts as bright lines with posterior shadowing.

Adductor Longus adductor-longus

Hip adduction

 **Patient Position** adductor-longus-position.jpg
Position the patient supine with the hip slightly abducted and externally rotated, and the knee flexed with the sole of the foot resting against the...

 **Probe + Needle Site** adductor-longus-probe.jpg
Patient supine, hip slightly abducted and externally rotated. Place probe transversely on proximal medial thigh, 3-5 cm below... Needle: Insert the needle perpendicular to the skin into the muscle belly just lateral to the palpable tendon.

 **Ultrasound Image** adductor-longus-us.jpg
Identify subcutaneous fat and the fascia lata as the superficial layers.

Adductor Magnus adductor-magnus

Hip adduction, Hip extension

 **Patient Position** adductor-magnus-position.jpg
Position the patient prone with the legs slightly apart, or supine with the hip abducted and externally rotated.

 **Probe + Needle Site** adductor-magnus-probe.jpg
Patient prone, legs slightly apart. Place probe transversely on posteromedial thigh at junction of middle and distal thirds.... Needle: Insert the needle from the posteromedial aspect of the thigh, angling slightly anteriorly.

 **Ultrasound Image** adductor-magnus-us.jpg
Identify subcutaneous fat and the deep fascia as the superficial layers.

Biceps Femoris biceps-femoris

Knee flexion, Hip extension

 **Patient Position** biceps-femoris-position.jpg
Position the patient prone with the knee slightly flexed over a pillow or bolster.

 **Probe + Needle Site** biceps-femoris-probe.jpg
Patient prone, knee flexed over pillow. Place probe transversely on lateral posterior thigh at junction of proximal and middle... Needle: Insert the needle perpendicular to the skin into the lateral muscle belly of the posterior thigh.

 **Ultrasound Image** biceps-femoris-us.jpg
Identify subcutaneous fat and the deep fascia (fascia lata) as the superficial layers.

Iliopsoas iliopsoas

Hip flexion

 **Patient Position** iliopsoas-position.jpg
Position the patient supine with the hip in neutral position and the knee extended.

 **Probe + Needle Site** iliopsoas-probe.jpg
Patient supine, hip neutral. Place curvilinear probe transversely on proximal anterior thigh, 2-3 cm below inguinal ligament,... Needle: Under ultrasound guidance, insert the needle lateral to the femoral vessels and advance toward the iliopsoas muscle belly.

 **Ultrasound Image** iliopsoas-us.jpg
Identify subcutaneous fat, the fascia lata, and the Sartorius muscle as the most superficial structures laterally.

Peroneus Longus peroneus-longus

Ankle eversion, Ankle plantarflexion

 **Patient Position** peroneus-longus-position.jpg
Position the patient supine with the leg internally rotated to expose the lateral compartment, or in a lateral decubitus position with the affected...

 **Probe + Needle Site** peroneus-longus-probe.jpg
Patient supine, leg internally rotated. Place probe transversely on lateral proximal leg, 5-8 cm below fibular head. Photo from... Needle: Insert the needle perpendicular to the skin into the lateral muscle belly, aiming toward the fibula.

 **Ultrasound Image** peroneus-longus-us.jpg
Identify subcutaneous fat and the crural fascia as the superficial layers.

Extensor Digitorum Longus ed1

Toe extension, Ankle dorsiflexion

 **Patient Position** ed1-position.jpg
Position the patient supine with the leg in neutral position and the foot relaxed.

 **Probe + Needle Site** ed1-probe.jpg
Patient supine, leg neutral. Place probe transversely on anterolateral proximal leg at proximal-middle third junction. Photo from... Needle: Insert the needle perpendicular to the skin into the anterolateral muscle belly.

 **Ultrasound Image** ed1-us.jpg
Identify subcutaneous fat and the crural fascia as the superficial layers.

Adductor Brevis adductor-brevis

Hip adduction (deep, short adductor)

Patient Position adductor-brevis-position.jpg

Position the patient supine with the hip abducted 30-45 degrees, knee flexed, and the leg externally rotated (frog-leg position).

Probe + Needle Site adductor-brevis-probe.jpg

Patient supine, frog-leg position. Place probe transversely on proximal medial thigh at the adductor longus origin. Photo from... Needle: USG is strongly recommended — insert the needle through the adductor longus to reach the adductor brevis layer deep to it.

Ultrasound Image adductor-brevis-us.jpg

Place the probe transversely on the proximal medial thigh with the hip abducted.

Gracilis gracilis

Hip adduction + knee flexion (crosses both joints)

Patient Position gracilis-position.jpg

Position the patient supine with the hip abducted 30-45 degrees and knee slightly flexed.

Probe + Needle Site gracilis-probe.jpg

Patient supine, frog-leg position. Place probe transversely on proximal medial thigh. Photo from medial view showing probe on... Needle: Insert the needle perpendicular to the skin on the medial thigh into the superficial muscle belly.

Ultrasound Image gracilis-us.jpg

Place the probe transversely on the proximal medial thigh with the hip abducted.

Foot Lumbricals foot-lumbricals

MTP flexion + IP extension in the foot (contributes to claw toe deformity)

Patient Position foot-lumbricals-position.jpg

Position the patient prone with the foot hanging off the bed, or supine with the ankle dorsiflexed to expose the sole.

Probe + Needle Site foot-lumbricals-probe.jpg

Patient prone or supine with foot exposed. Place hockey-stick probe longitudinally on plantar surface along the metatarsal. Photo... Needle: Insert the needle from the plantar surface perpendicular to the skin, just proximal to the MTP joint of the target toe, on the...

Ultrasound Image foot-lumbricals-us.jpg

Place the probe longitudinally on the plantar surface along the metatarsal of the target toe.

Sternocleidomastoid (SCM) scm

Head rotation to opposite side, ipsilateral lateral flexion

Patient Position scm-position.jpg

Position the patient supine with the head turned slightly to the contralateral side to make the SCM prominent but not maximally stretched.

Probe + Needle Site scm-probe.jpg

Patient supine, head turned slightly contralateral. Place probe transversely on lateral neck at thyroid cartilage level. Photo... Needle: Palpate the muscle belly at the mid-cervical level (approximately the level of the thyroid cartilage). Grasp the muscle between...

Ultrasound Image scm-us.jpg

Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

Upper Trapezius upper-trapezius

Scapular elevation, head lateral flexion

Patient Position upper-trapezius-position.jpg

Position the patient seated upright with arms relaxed at the sides, or prone with the forehead resting on hands.

Probe + Needle Site upper-trapezius-probe.jpg

Patient seated, arms relaxed. Place probe transversely on upper trapezius ridge, midway between cervical spine and acromion.... Needle: Insert the needle perpendicular to the skin surface into the midpoint of the palpated muscle bulk, advancing 1-2 cm into the...

Ultrasound Image upper-trapezius-us.jpg

Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

Levator Scapulae levator-scapulae

Scapular elevation, cervical lateral flexion and rotation

Patient Position levator-scapulae-position.jpg

Position the patient seated upright with arms relaxed, or prone with the forehead resting on hands. Slight contralateral lateral flexion of the neck...

Probe + Needle Site levator-scapulae-probe.jpg

Patient seated, arms relaxed. Place probe transversely over superior angle of scapula, 2-3 cm lateral to C7-T2 spinous processes.... Needle: Identify the superior angle of the scapula by palpating the superomedial corner of the scapula — this is the most reliable...

Ultrasound Image levator-scapulae-us.jpg

Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

Splenius Capitis splenius-capitis

Ipsilateral head rotation and extension

Patient Position splenius-capitis-position.jpg

Position the patient seated upright with the neck slightly flexed to open the posterior cervical space, or prone with the forehead resting on hands.

Probe + Needle Site splenius-capitis-probe.jpg

Patient seated, neck slightly flexed. Place probe transversely on posterolateral neck at C2-C4 level, 2-3 cm lateral to midline.... Needle: Insert the needle perpendicular to the skin, advancing through the trapezius (1-1.5 cm) and into the splenius capitis beneath it...

Ultrasound Image splenius-capitis-us.jpg

Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

Semispinalis Capitis semispinalis-capitis

Bilateral head and neck extension (retrocollis)

Patient Position semispinalis-capitis-position.jpg

Position the patient seated with the neck slightly flexed to open the posterior cervical space, or prone with the forehead resting on hands.

Probe + Needle Site semispinalis-capitis-probe.jpg

Patient seated, neck flexed. Place probe transversely on posterior neck at C2-C5 level, 1-2 cm lateral to midline. Photo from... Needle: This is a DEEP muscle — the needle must traverse the trapezius and splenius capitis to reach it. USG guidance is essential to...

Ultrasound Image semispinalis-capitis-us.jpg

Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

Scalenes (Anterior, Middle, Posterior) scalenes

Ipsilateral lateral neck flexion (laterocollis), accessory inspiration

Patient Position scalenes-position.jpg

Position the patient supine with the head in neutral position or turned slightly to the contralateral side to expose the lateral neck.

Probe + Needle Site scalenes-probe.jpg

Patient supine, head neutral or slightly turned contralateral. Place probe transversely on lateral neck at C5-C7 level, posterior... Needle: Under ultrasound, identify the target scalene muscle and confirm the location of the brachial plexus, subclavian/vertebral...

Ultrasound Image scalenes-us.jpg

Identify the SCM as the most superficial anterolateral muscle — use it as your anterior landmark.

Longissimus Capitis longissimus-capitis

Head and neck extension (retrocollis), ipsilateral lateral flexion and rotation

Patient Position longissimus-capitis-position.jpg

Position the patient seated with the neck slightly flexed to open the posterior cervical space, or prone with the forehead resting on hands.

Probe + Needle Site longissimus-capitis-probe.jpg

Patient seated, neck slightly flexed. Place probe transversely on posterolateral neck at C4-C7 level, 2-4 cm lateral to midline.... Needle: This is a DEEP muscle in the erector spinae group — the needle must traverse the trapezius and splenius capitis to reach it. USG...

Ultrasound Image longissimus-capitis-us.jpg

Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

Masseter masseter

Jaw clenching, bruxism, oromandibular dystonia (jaw closing type)

☐ Patient Position masseter-position.jpg

Position the patient seated upright or supine with the jaw relaxed and slightly open.

☐ Probe + Needle Site masseter-probe.jpg

Patient seated upright with jaw relaxed. Place probe transversely on lateral face over the mandibular ramus midway between... Needle: Ask the patient to clench their teeth forcefully — the masseter will bulge prominently as a thick rectangular mass over the...

☐ Ultrasound Image masseter-us.jpg

Place the probe transversely on the lateral face over the mandibular ramus, midway between the zygomatic arch and the angle of... Depth ≈ 2.0-3.0 cm.

Temporalis temporalis

Jaw clenching, bruxism, oromandibular dystonia (jaw closing type), temporal headache

☐ Patient Position temporalis-position.jpg

Position the patient seated upright or supine with the jaw relaxed.

☐ Probe + Needle Site temporalis-probe.jpg

Patient seated upright with jaw relaxed. Place probe transversely in the temporal fossa above the zygomatic arch. Photo from... Needle: Insert the needle perpendicular to the skull surface into the muscle belly at each site. Advance only 0.5-1 cm — the temporalis...

☐ Ultrasound Image temporalis-us.jpg

Place the probe transversely on the temporal fossa above the zygomatic arch. Apply minimal pressure — the temporalis is thin and... Depth ≈ 1.0-1.5 cm.

Lateral Pterygoid lateral-pterygoid not typically US-guided

Jaw opening dystonia, jaw protrusion, oromandibular dystonia (jaw opening type), jaw deviation to contralateral side

☐ Patient Position lateral-pterygoid-position.jpg

Position the patient seated upright or supine with the jaw slightly open (to translate the condyle forward and open the coronoid notch).

☐ Probe + Needle Site lateral-pterygoid-probe.jpg

EMG-guided injection — no ultrasound probe placement. Patient seated with jaw slightly open. Needle inserted through coronoid... Needle: Insert a 25-27G, 1.5-2 inch needle through the coronoid notch, directing it medially and slightly anteriorly toward the... No US — red dot only; blue bar optional.

☐ Ultrasound Image lateral-pterygoid-us.jpg

Optional — not typically ultrasound-guided.

Medial Pterygoid medial-pterygoid

Jaw clenching, oromandibular dystonia (jaw closing type), synergist with masseter for jaw elevation

☐ Patient Position medial-pterygoid-position.jpg

Position the patient seated upright or supine with the jaw slightly open.

☐ Probe + Needle Site medial-pterygoid-probe.jpg

Patient seated with jaw slightly open. For US guidance: place probe in submandibular region below mandibular angle, angled... Needle: SUBMANDIBULAR (EXTRAORAL) APPROACH (preferred for US guidance): Palpate the medial surface of the mandibular angle from the...

☐ Ultrasound Image medial-pterygoid-us.jpg

Place the probe in the submandibular region below and medial to the angle of the mandible, angled superiorly. Depth ≈ 2.5-4.0 cm.

Orbicularis Oculi orbicularis-oculi not typically US-guided

Blepharospasm (involuntary forceful eye closure), hemifacial spasm

☐ Patient Position orbicularis-oculi-position.jpg

Position the patient seated upright in a well-lit room. The patient should be relaxed with eyes gently closed or in their resting blepharospasm...

☐ Probe + Needle Site orbicularis-oculi-probe.jpg

Landmark-guided injection — no ultrasound used. Patient seated upright in well-lit room. Six injection sites distributed around... Needle: Insert the needle subcutaneously (bevel up) at each site, directed AWAY from the globe. The muscle is extremely thin (1-2 mm) and... No US — red dot only; blue bar optional.

☐ Ultrasound Image orbicularis-oculi-us.jpg

Optional — not typically ultrasound-guided.

Corrugator Supercilii corrugator-supercilii not typically US-guided

Blepharospasm (glabellar component), brow furrowing, glabellar dystonia

☐ Patient Position corrugator-supercilii-position.jpg

Position the patient seated upright. Ask them to frown forcefully — the corrugator creates the vertical (11 lines) furrows between the brows.

☐ Probe + Needle Site corrugator-supercilii-probe.jpg

Landmark-guided injection — no ultrasound used. Patient seated upright, frowning to activate the corrugator. Injection at the... Needle: Insert a 30G needle at a shallow angle (20-30 degrees from the skin surface) directed laterally along the corrugator muscle... No US — red dot only; blue bar optional.

☐ Ultrasound Image corrugator-supercilii-us.jpg

Optional — not typically ultrasound-guided.

Procerus procerus not typically US-guided

Blepharospasm (glabellar component), horizontal nasal root furrows, glabellar dystonia

☐ Patient Position procerus-position.jpg

Position the patient seated upright. Ask them to 'scrunch' their nose or pull their brows down toward the nose — the procerus creates horizontal...

☐ Probe + Needle Site procerus-probe.jpg

Landmark-guided injection — no ultrasound used. Patient seated upright, actively scrunching the nose to identify the procerus.... Needle: Insert a 30G needle perpendicular to the skin or at a shallow angle into the midline nasal root/glabella junction, advancing only... No US — red dot only; blue bar optional.

☐ Ultrasound Image procerus-us.jpg

Optional — not typically ultrasound-guided.

Frontalis frontalis not typically US-guided

Brow elevation dystonia, forehead spasm, involuntary brow raising; NOTE: this is primarily a cosmetic target (horizontal forehead lines) and is NOT a typical spasticity target

☐ Patient Position frontalis-position.jpg

Position the patient seated upright. Ask them to raise their eyebrows forcefully — the frontalis will contract, creating horizontal forehead lines.

☐ Probe + Needle Site frontalis-probe.jpg

Landmark-guided injection — no ultrasound used. Patient seated upright, raising eyebrows to activate frontalis. Four to five... Needle: Standard cosmetic approach: inject at 4-5 sites across the mid-forehead, each approximately 1.5-2 cm above the brow and spaced... No US — red dot only; blue bar optional.

☐ Ultrasound Image frontalis-us.jpg

Optional — not typically ultrasound-guided.

Platysma platysma not typically US-guided

Anterocollis (forward head flexion), neck banding, platysmal dystonia, involuntary neck tightening

 Patient Position platysma-position.jpg

Position the patient seated upright or supine. Ask them to grimace or depress the mandible with lips retracted to make the platysmal bands prominent.

 Probe + Needle Site platysma-probe.jpg

Landmark-guided injection — platysmal bands are identified visually by asking the patient to grimace. Patient seated upright,... Needle: Insert a 30G needle subcutaneously into each platysmal band, directing it along the length of the band. Advance only 2-3 mm — the... No US — red dot only; blue bar optional.

 Ultrasound Image platysma-us.jpg

Optional — not typically ultrasound-guided.

Mentalis mentalis not typically US-guided

Chin dimpling dystonia, involuntary chin puckering, mental crease deepening, lower lip protrusion

 Patient Position mentalis-position.jpg

Position the patient seated upright. Ask them to 'pout' or push the lower lip forward — the mentalis will contract, producing the characteristic...

 Probe + Needle Site mentalis-probe.jpg

Landmark-guided injection — no ultrasound used. Patient seated upright, pouting to activate the mentalis and display chin... Needle: For unilateral dystonia or dimpling, inject the affected side only. For bilateral chin dimpling, inject both sides. No US — red dot only; blue bar optional.

 Ultrasound Image mentalis-us.jpg

Optional — not typically ultrasound-guided.